Impression Forecast Model

Group Members

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Academic Context

CSCI-635: Foundations of Machine Learning

Rochester Institute of Technology

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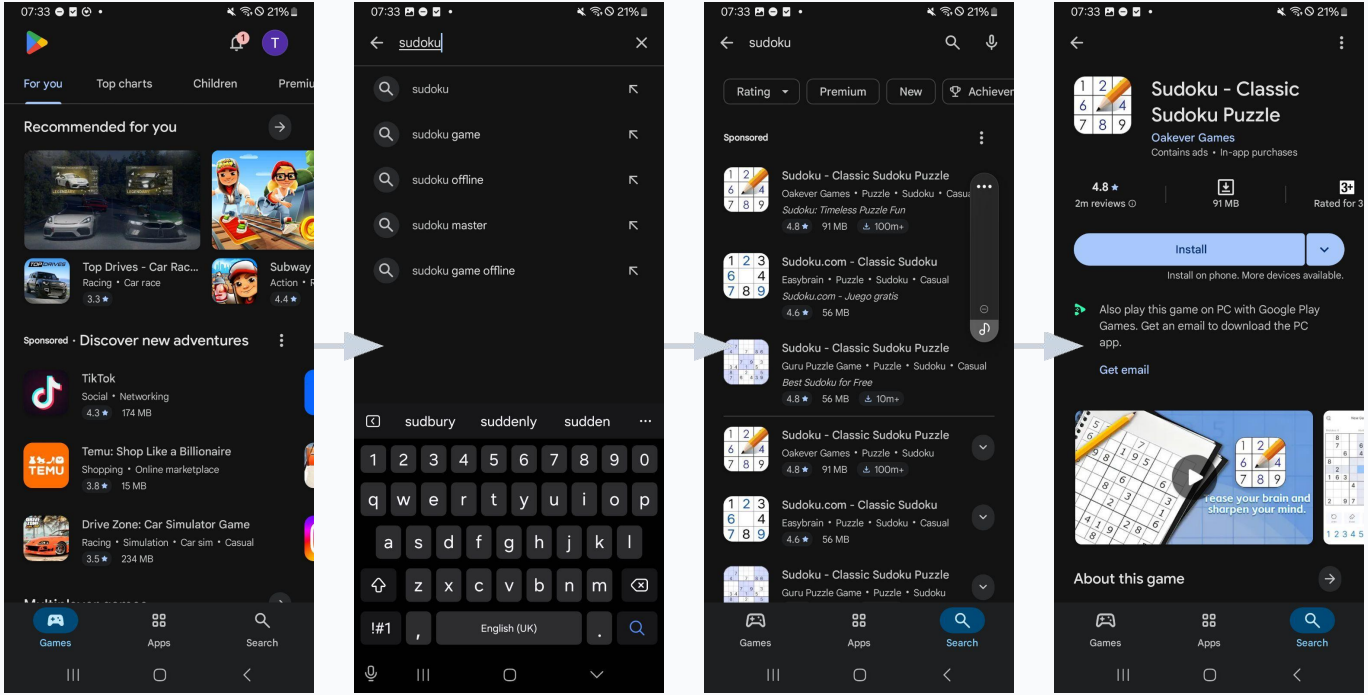
What is an Impression

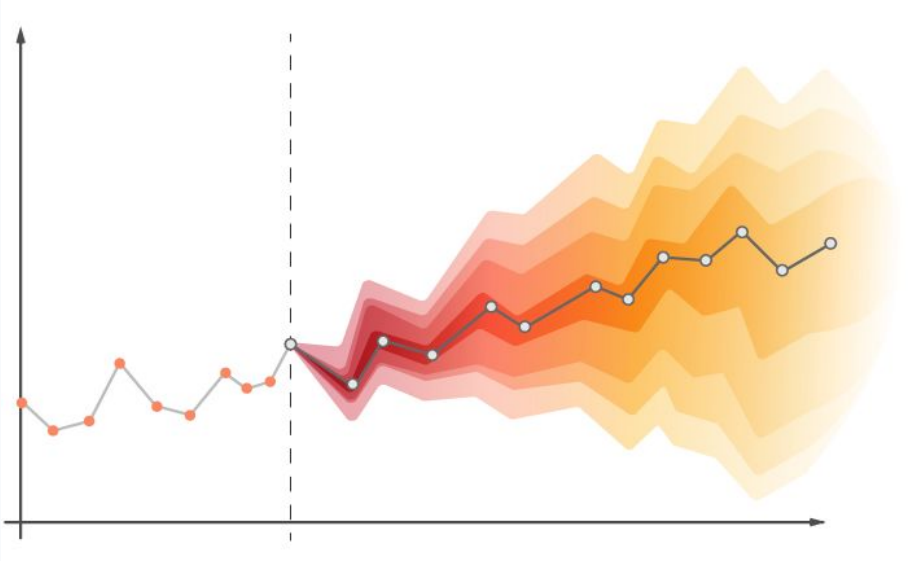


STEPS

1. Browse.
2. Search
3. Select
4. Install.

What is forecasting

Predicting Future outcome based on historical data



https://medium.com/analytics-vidhya/time-series-forecasting-c73dec0b7533

Impression / Downloads Forecast

Key Use Cases

- Setting Goals and Tracking
- Financial Planning
- Product update planning
- Seasonal planning for holidays
- Outlier Detection

Problem Statement

Objective:

*"To develop a time-series regression model that predicts the number of downloads for an application on day **T**, given a feature vector **X** representing the preceding **N** days of historical data."*

f(Xt-n, ..., Xt-1) Yt

Dataset

Data Source
Proprietary data from Google Play Console, available to developers publishing on the platform.

Target App
Car Driving School.

Collection Pipeline
Check our repo for more info.

Data Preprocessing

Missing Rows

Handling incomplete data entries to ensure dataset continuity.

Null Fixes

Implementing strategies like interpolation or mean substitution for null values.

Add weekend column

Feature engineering to capture weekly seasonal patterns in the time series.

Methodology

Models used in project

Traditional Models

AutoRegressive Integrated Moving Average

The fundamental statistical framework for time series forecasting.

ARIMA Family

ARIMA

Standard non-seasonal model

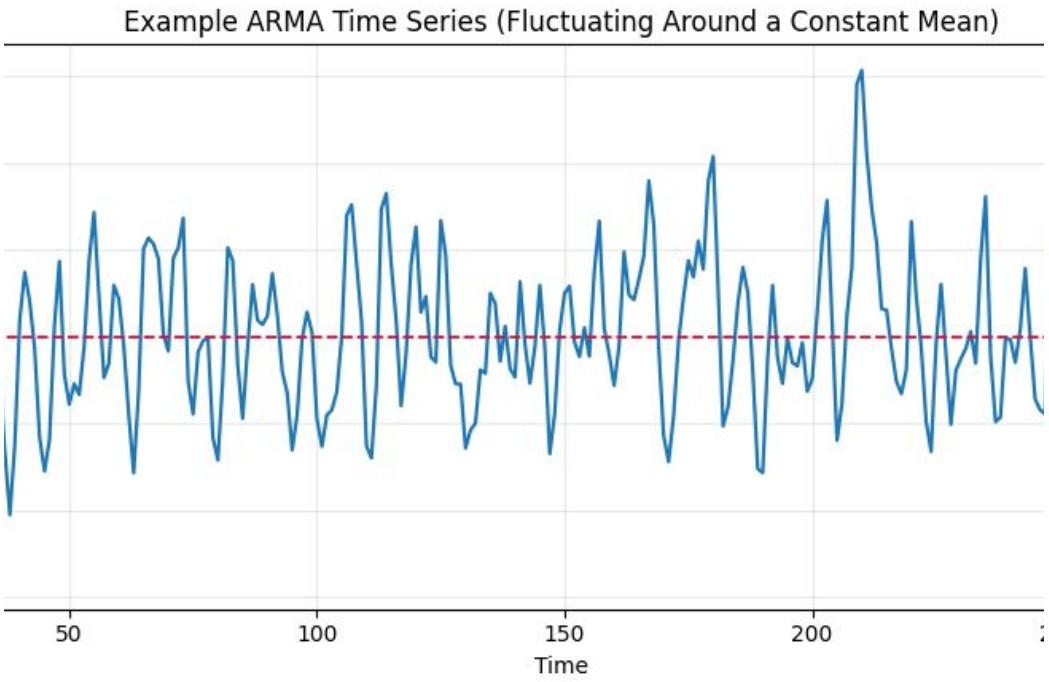
SARIMA

Handles seasonality in data

SARIMAX

Includes exogenous variables

A little Background - ARMA



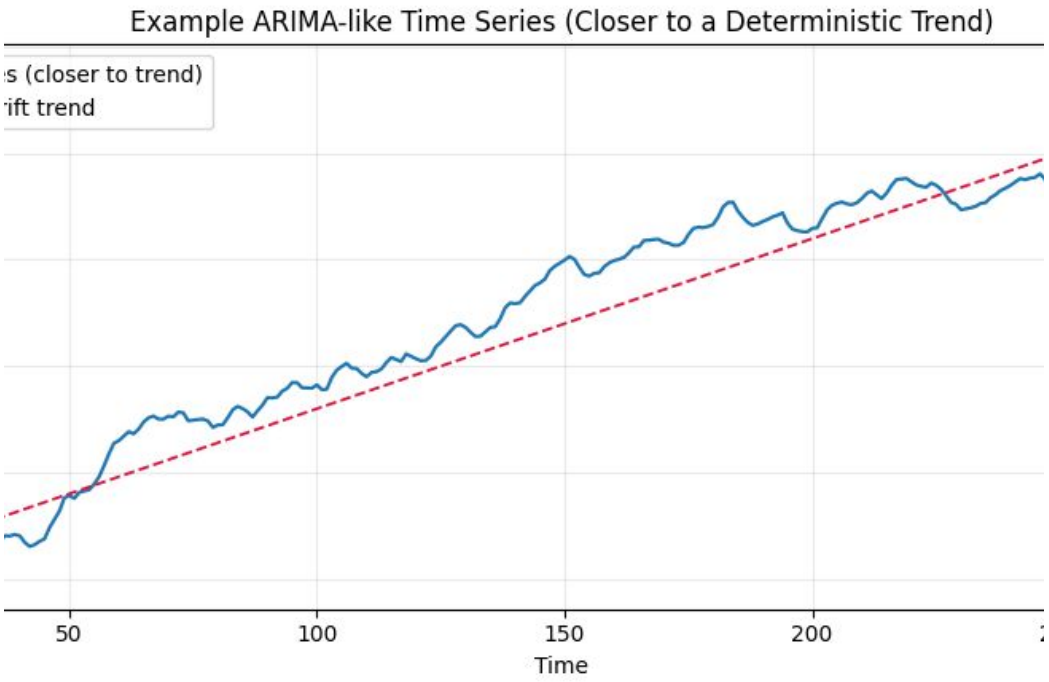
AR - Autoregressive

Uses the dependency between an observation and a number of lagged observations.

MA - Moving Average

Uses the dependency between an observation and a residual error from a moving average model.

ARIMA



AR - Autoregressive

Uses the dependency between an observation and a number of lagged observations.

I - Integrated

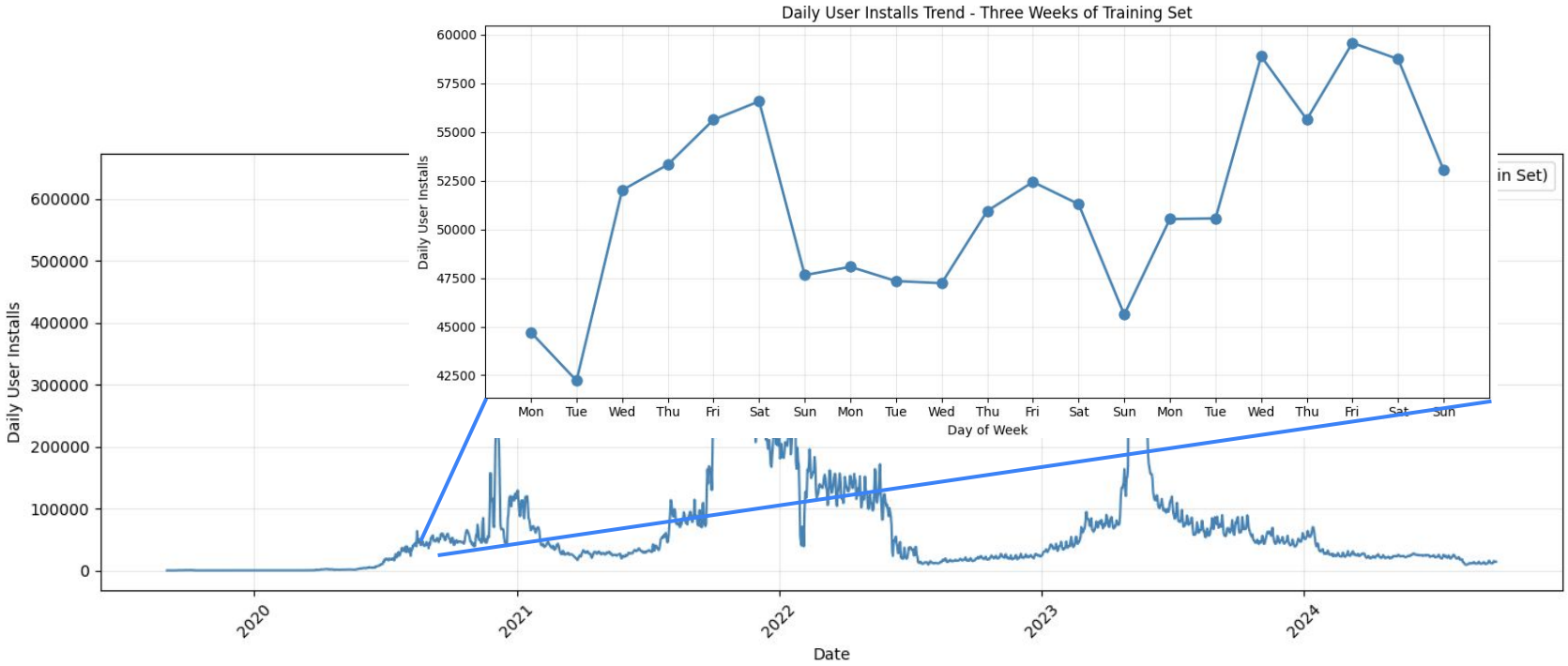
Uses differencing of raw observations in order to make the time series stationary.

MA - Moving Average

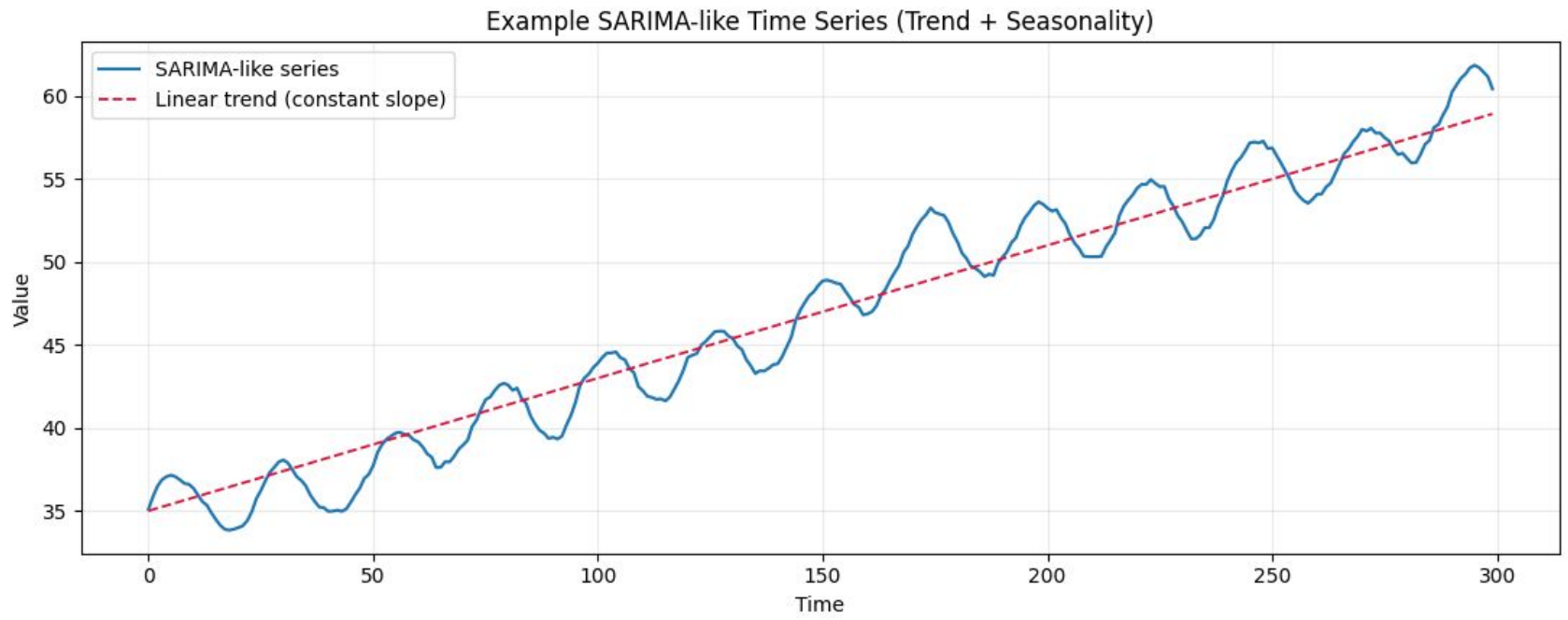
Uses the dependency between an observation and a residual error from a moving average model.

We Can Still Do Better...

Let’s Zoom In


Daily User Installs Trend - Three Weeks of Training Set
Daily User Installs
60000
57500
55000
52500
50000
47500
45000
42500
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Day of Week
in Set)
600000
500000
400000
300000
200000
100000
0
Daily User Installs
2020
2021
2022
2023
2024
Date

Seasonality with ARIMA - SARIMA



One Final Piece - SARIMAX

Exogenous Variables

Just SARIMA but with multiple features.

LSTM

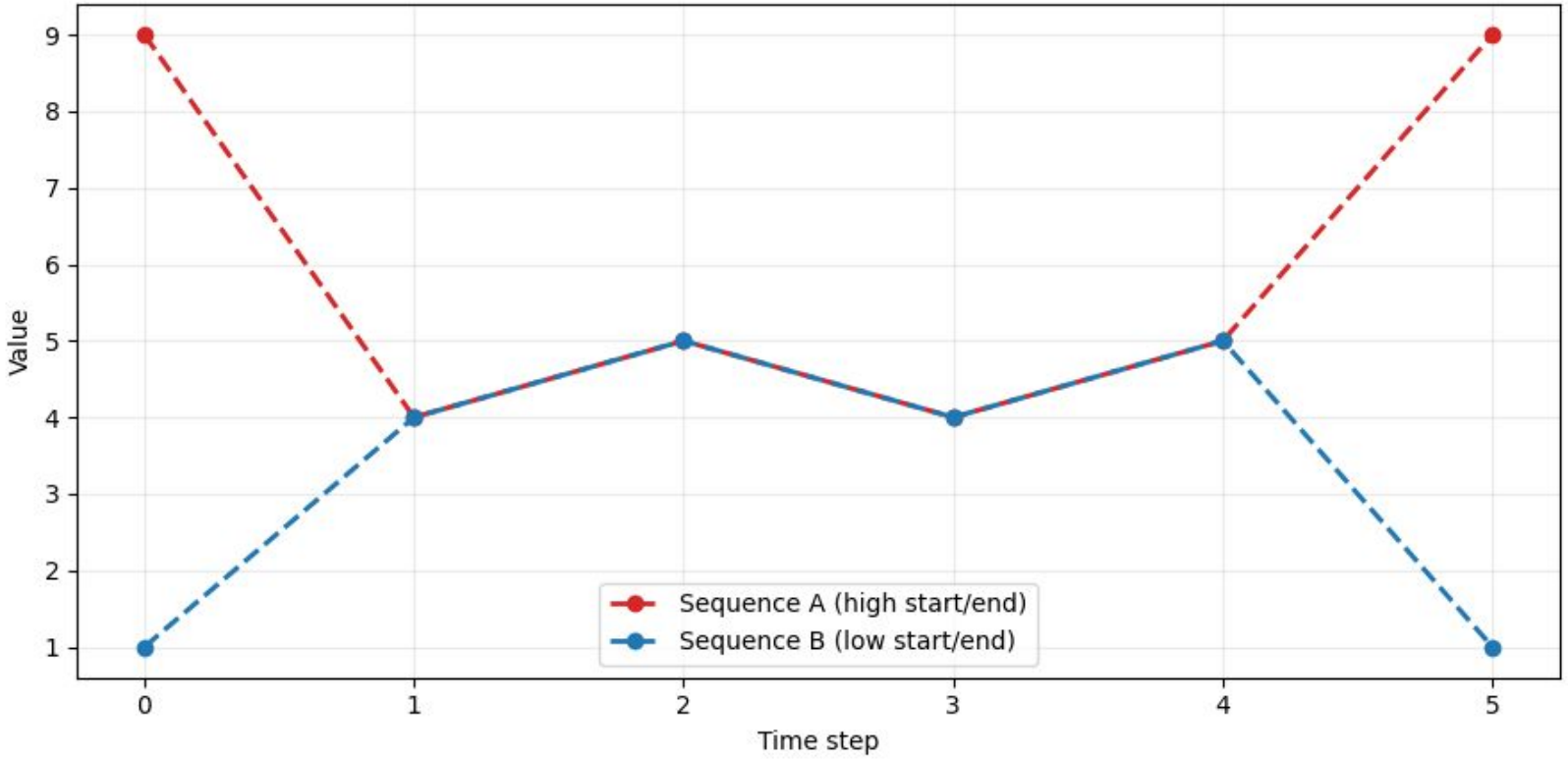
Long Short Term Memory

A specialized RNN architecture designed to handle long-range dependencies in sequential data.

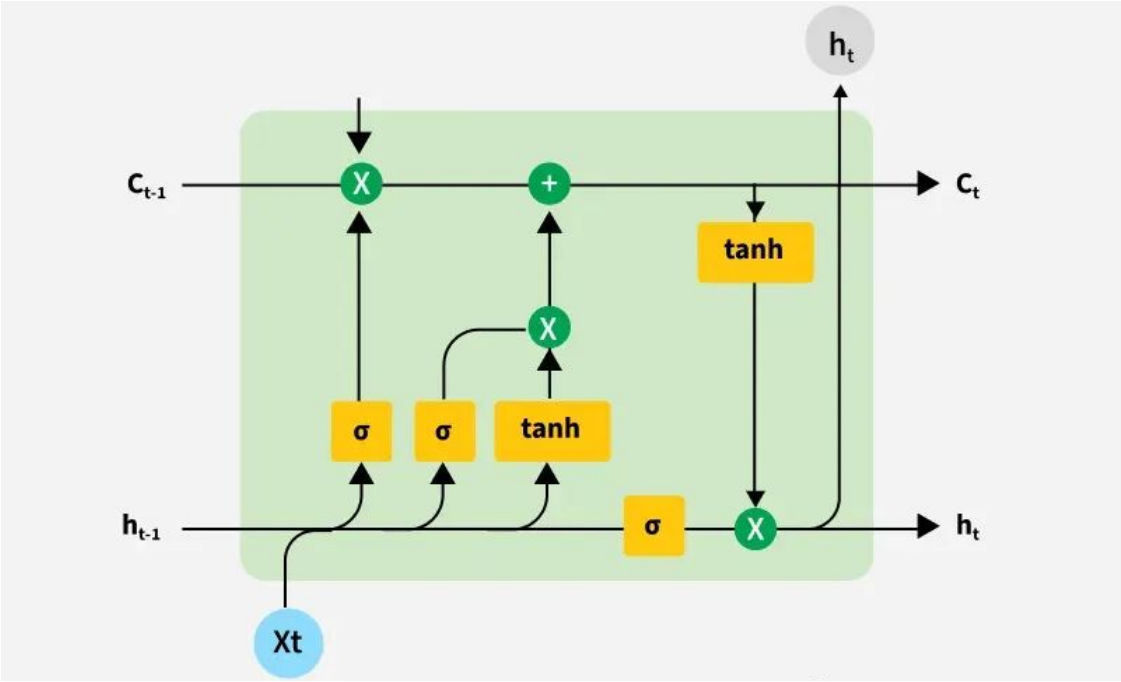
Dual Memory Mechanism

Memorizes important long-term patterns alongside short-term trends.

Why Long Term Memory Matters?



LSTM Internal Gating Mechanism

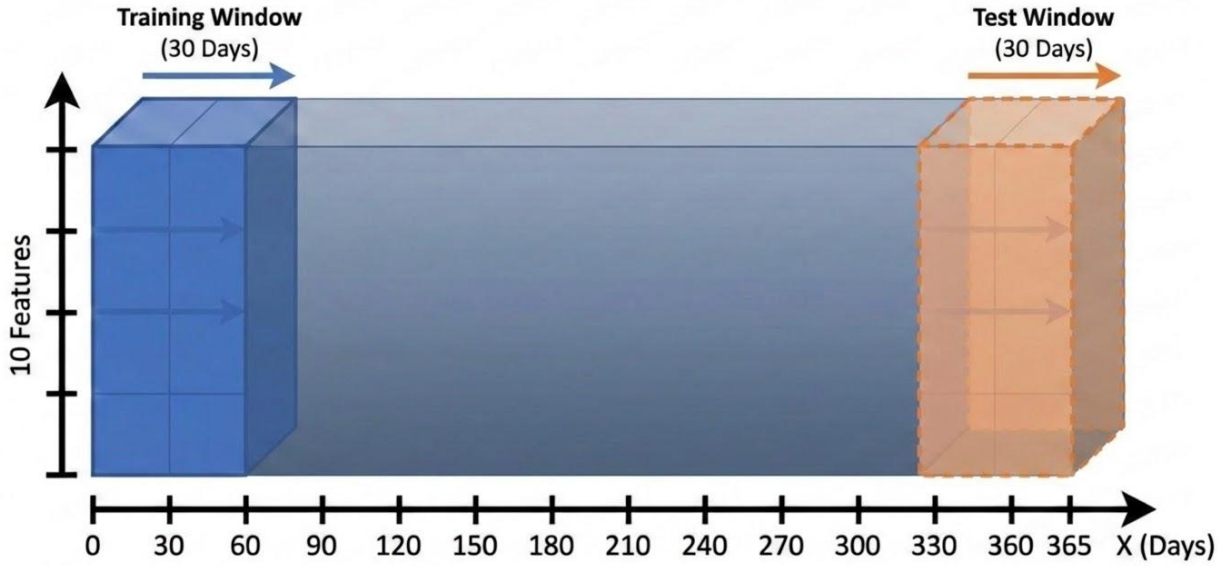


Core Components

- **Cell State (C):** The "long-term memory" conveyor belt.
- **Hidden State (h):** The "short-term memory" output.
- **Gates:** Regulate flow via Sigmoid (σ) and Tanh functions.

Source: GeeksForGeeks

How We Are Using The Dataset

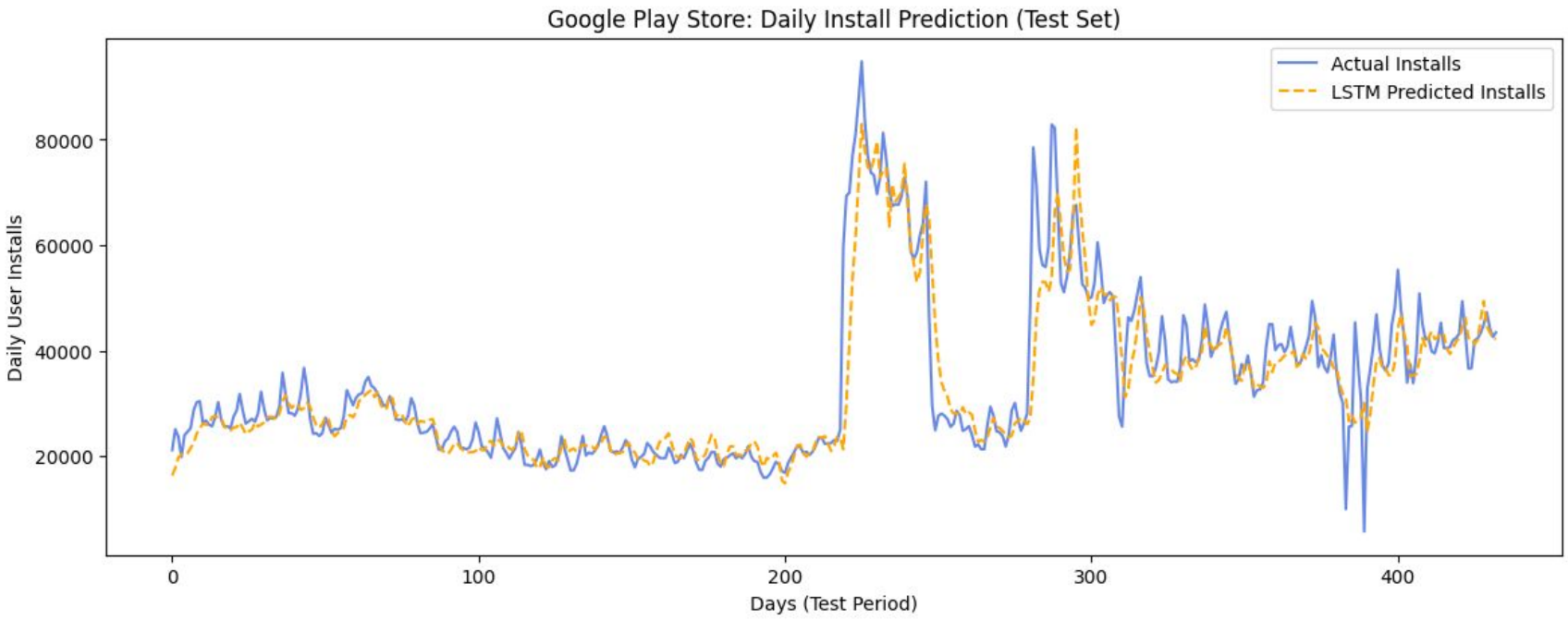


But What About Seasonality?

Addressing Periodic Patterns

Turns out creating a new feature as **weekend** does the trick for **LSTM**.

Comparison of Actual And Predicted Installs (LSTM)



Google Play Store: Daily Install Prediction (Test Set)

State of the Art

State-of-the-Art Models

"Can a model trained on your data beat one trained on the entire internet?"

Facebook's Prophet

The Classical Benchmark

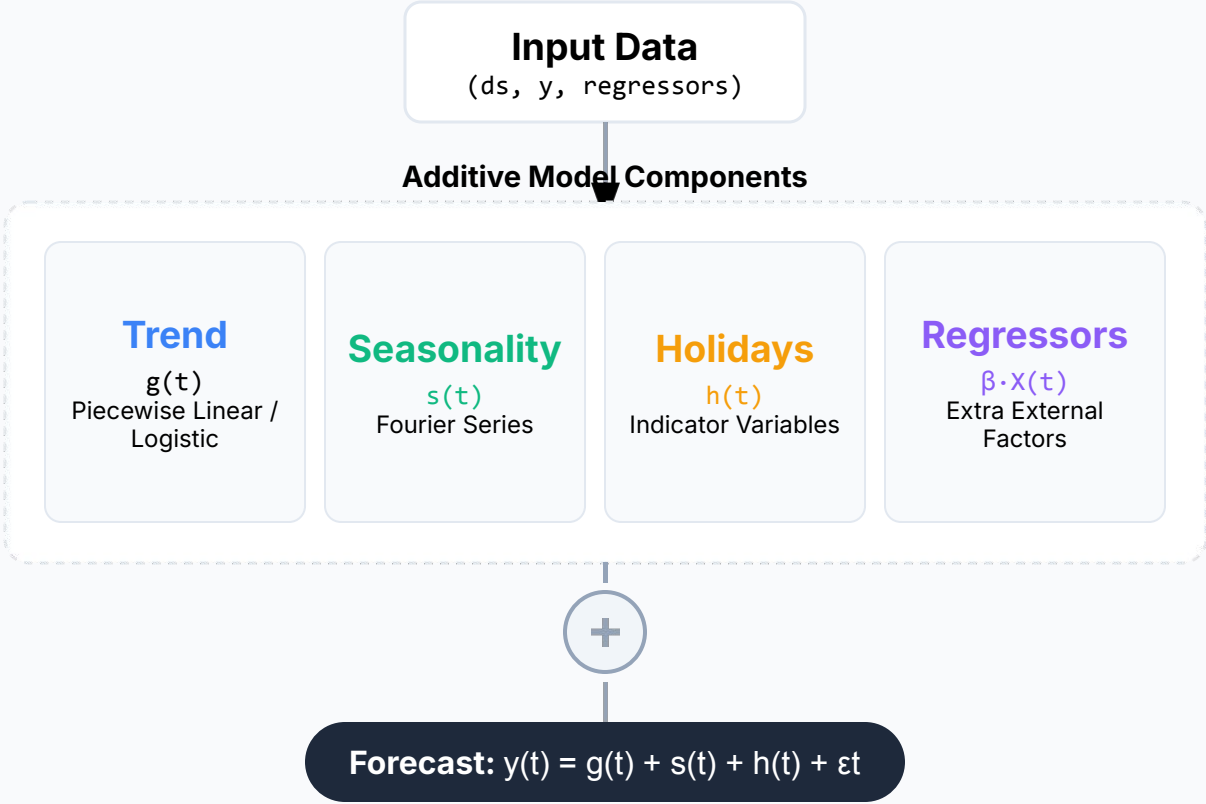
- **Type:** Additive Regression Model
- **Innovation:** Domain knowledge, holidays, seasonality, regressors
- **Key Difference:** Trained on your specific app data with 8 external features

Google's TimesFM

The Modern Foundation

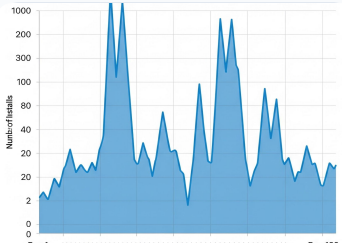
- **Type:** Transformer-based Foundation Model
- **Innovation:** Pretrained on 100B real-world time points globally
- **Key Difference:** Zero-shot forecasting, no training needed

How Prophet works?



TimesFM Architecture

Long-term Time Series Forecasting (Patch-based)



Input History

128-day historical data window

`Context Length: 128`

Patching

Split into four 32-day patches

`Patch Size: 32`

Transformer Encoder

Attention Mechanism & Temporal Pattern Extraction

Forecast Output

Projects forecast for next N days

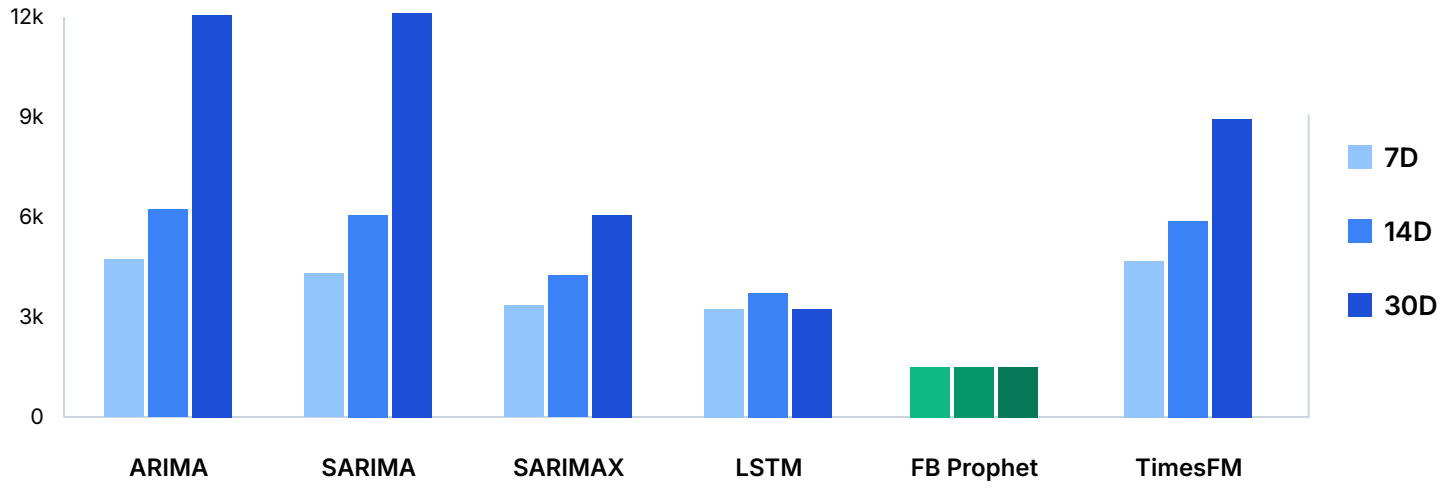
`Horizon: 7, 14, 30`

Results

Performance Evaluation & Model Metrics

Model Performance Comparison

MAE (Mean Absolute Error) metrics across different forecast horizons



Conclusions

Analysis and Future Work

Model Comparison Analysis

ARIMA

TYPE
- Traditional ML

STRENGTHS
- Interpretable
- External Regressor

WEAKNESSES:
- Higher Loss
- Longer Training

LSTM

TYPE
- Deep Learning

STRENGTHS
- Good Results
- External Regressor

WEAKNESSES:
- Not Interpretable
- Longer Training

Prophet

TYPE
- Curve Fitting

STRENGTHS
- Best Results
- External Regressor

WEAKNESSES:
- Prone to Overfitting

TimesFM

TYPE
- Foundation Model

STRENGTHS
- Zero Training

WEAKNESSES:
- Higher Loss
- Not Interpretable

Conclusions & Next Steps

Limitations

- Only one data source
- Limited feature set in initial modeling

Future Works

- **Model Generalization:** Predict from other datasets using pretrained models
- **Robustness Check:** Evaluate underfit/overfit on diverse game data
- **Productization:** Create tool for data import and forecast projection for other developers
- **Other Models:** ETS, SVR, Autoformer, Amazon Chronos, TimeGPT

Questions & Discussion

Github Repo

Scan to access the
Project Repository

Thank you for your attention.

GITHUB REPOSITORY

github.com/Junaidsgs/IFM